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EXPERIMENT 10

Pattern Storage of 10 Digits using a Discrete Hopfield Network

1. Aim

To implement a Discrete Hopfield Network in MATLAB for storing and recalling patterns representing the 10 decimal digits (0–9), and to test the network's ability to recover a digit from a noisy or incomplete input pattern.

3. Software Requirements

- MATLAB
 - Computer system with Windows/Linux/macOS
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4. Theory

A Hopfield Network is a recurrent neural network primarily used as an associative memory. It stores patterns in the connections between neurons and can retrieve a stored pattern when a noisy or partially incorrect version of that pattern is provided as input.

In a Discrete Hopfield Network, the neuron states are represented using bipolar values, **+1** and **-1**. Therefore, an active pixel represented by binary **1** is converted to **+1**, while an inactive pixel represented by binary **0** is converted to **-1**.

In this experiment, each decimal digit from 0 to 9 is represented using a 7×5 grid. Therefore, each digit contains:

$$7 \times 5 = 35$$

pixels, which correspond to 35 neurons in the Hopfield Network.

The ten digit patterns are converted into bipolar vectors and stored in the network using the Hebbian learning rule. For multiple stored patterns, the weight matrix is calculated as:

$$W = \sum_k X_k X_k^T$$

where X_k represents the bipolar vector corresponding to the k^{th} stored digit.

The diagonal elements of the weight matrix are set to zero to eliminate self-connections between neurons.

During recall, a selected digit is corrupted by introducing noise into some of its pixels. The noisy pattern is then given to the Hopfield Network.

Each neuron calculates its local field using the weighted sum of the current neuron states:

$$Net_i = \sum_j W_{ij} x_j$$

The neuron state is then updated according to the sign of the net input:

$$x_i(\text{new}) = \begin{cases} +1, & Net_i \geq 0 \\ -1, & Net_i < 0 \end{cases}$$

The neurons are updated repeatedly until the network reaches a stable state or the maximum number of iterations is reached.

Finally, the recalled pattern is compared with all stored patterns using dot-product similarity. The stored digit having the highest similarity is selected as the recalled digit.

5. Digit Pattern Representation

Each digit is represented using a 7×5 binary grid.

For example, the digit 8 is represented as:

0 1 1 1 0

1 0 0 0 1

1 0 0 0 1

0 1 1 1 0

1 0 0 0 1

1 0 0 0 1

0 1 1 1 0

The binary values are converted to bipolar values:

0 $\rightarrow$ -1

1 $\rightarrow$ +1

Therefore, every digit becomes a vector containing 35 bipolar values.

7. Algorithm

1. Start MATLAB.
2. Define ten 7×5 patterns representing digits 0–9.
3. Convert all binary patterns into bipolar form.
4. Store the patterns in a matrix PP.
5. Calculate the Hopfield weight matrix using $W=PPTW=PP^T$.
6. Set the diagonal elements of WW to zero.
7. Select digit 8 as the test digit.
8. Introduce 15% noise into the selected pattern.
9. Initialize the network with the noisy pattern.
10. Update the 35 neurons iteratively using the weighted input.
11. Stop when the network becomes stable or reaches 20 iterations.
12. Calculate similarity between the recalled pattern and all stored patterns.
13. Select the pattern with maximum similarity.
14. Display the original, noisy, and recalled digits.
15. Stop.

11. Conclusion

The experiment demonstrates that a Discrete Hopfield Network can store multiple digit patterns and retrieve a stored pattern from a noisy input.

In this experiment, the digit 8 was selected, corrupted with noise, and supplied to the network. The Hopfield Network iteratively updated its neuron states and converged toward the corresponding stored pattern.